BibUpdate

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Version 1.4.0-r0

Automatically update the entries of a bibtex file

Usage:

```
bibupdate [-h|-H] [-a] [-X] [-c] [-d] [-k] [-i FIELDS] [-l LOG]
          [-m | -M | -n] [-o] [-q] [--threads N] [-w LEN]
          [--duplicates | --duplicate-papers | --duplicate-keys]
          [--list-duplicates | --list-papers | --list-keys]
          bibtexfile [outputfile]
```

This is a command line tool for updating the entries in a BibTeX file using the free tools provided by the American Mathematical Society: mrlookup, MathSciNet and MRef. By default bibupdate uses mrlookup to update `article`, `book`, `inproceedings` and `incollection` entries that have no `mrnumber` field. An `mrnumber` field, even an empty one, skips this lookup unless `--all` or `--check` is used. In addition, with the `-X` option, bibupdate can query the arXiv and add `eprint` and `archiveprefix` fields to any entry that it finds there.

Options by task:

Updating and checking

```
-m, --mrlookup        use mrlookup to update bibtex entries (default)
-M, --mathscinet      use mathscinet to update bibtex entries (less flexible)
-n, --mref            use mref to update bibtex entries
-a, --all             update or validate ALL BibTeX entries
-X, --arxiv           also add arXiv eprint fields to entries
-c, --check           check all entries and report duplicates
--threads N          number of concurrent AMS lookups (default: 1)
```

Duplicate listing

```
-d, --list-duplicates list duplicate papers and keys without writing a file
--list-papers        list duplicate papers only
--list-keys          list repeated citation keys only
```

Duplicate cleanup

```
--duplicates         remove duplicate papers and rename repeated keys
--duplicate-papers   remove duplicate papers, keeping the first
--duplicate-keys     rename repeated keys using authors and year
```

Output and formatting

```
-o, --overwrite       overwrite existing bibtex file
-i FIELDS, --ignored-fields FIELDS
                      a string of bibtex fields to ignore
-k, --keep_fonts      do NOT replace fonts \Bbb, \germ and \scr in titles
-w LEN, --wrap LEN    wrap bibtex fields to specified width
```

Reports and logging

```
-q, --quiet           brief updates and missed lookups (repeat to suppress)
-l LOG, --log LOG     log messages to specified file (defaults to stdout)
```

Help and version

```
-h, --help            show grouped options and exit
-H, --Help            print full program description
-v, --version         print the program version and exit
```

Note: As described below, you should check the new file for errors before deleting the original version of your BibTeX file.

By default, bibupdate does not change your original database file. Instead, it creates a new file with the name *updated_file.bib*, if your original file was *file.bib*. It is also possible to have it replace your current file (use carefully!), or to specify an output file.

BibTeX is widely used by the LaTeX community to maintain publication databases. This script attempts to add missing fields to the papers in a BibTeX database file by querying mrlookup and getting the missing information from there. This is not completely routine because to search on mrlookup you need either the authors or the title of the article and both of these can have non-standard representations. If the article is already published then it is also possible to use the publication year and its page numbers. To search on mrlookup we:

- use the first author's surname (accents and compound surnames can be problematic)
- use numeric page ranges or a single numeric page, when available for a published entry
- include the year for books, or when a numeric first page is available
- use the cleaned title when no numeric first page is available

The author query uses only the first author's surname: mrlookup does not interpret a BibTeX list of authors joined by `and` as separate names. Braces used to protect capitals in titles (for example `{Kac-Moody}`) are removed for searching and matching; the title stored in the entry retains its braces.

If there is a unique match from mrlookup then bibupdate updates the fields supplied by mrlookup, except for the citation key and the fields excluded by `--ignored-fields`. Local fields that are not supplied by mrlookup, such as `eprint`, are retained. By default, a message is printed whenever existing fields in the database are changed. If the title of the retrieved paper does not (fuzzily) match that of the original article then the entry is NOT updated and a warning message is printed.

Although some care is taken to make sure that the new BibTeX entries correspond to the same paper that the original entry referred to there is always a (small?) chance the new entry corresponds to an entirely different paper. In my experience this happens rarely, and mostly with unpublished manuscripts. In any case, before you delete your original BibTeX file *you are strongly advised to check the updated file BibTeX file carefully for errors!*

By default the program prints each updated entry's citation key on its own line, followed by an indented bibliography-style reference. Report lines wrap at 78 columns, preserving the complete authors, title and publication details. If arXiv fields are added as well, they appear in the same reference. `--check` uses this compact layout to show entries that would change, without writing an updated file. You can also compare the original and updated files using *diff*.

As bibupdate calls mrlookup this program will only be useful if you have papers in your database that are listed in MathSciNet. As described below it is also possible to call MathSciNet directly, however, this is less flexible because the `mrnumber` field for each paper is required.

I wrote this script because I wanted to automatically add links to journals, the arXiv and DOIs to the bibliographies of my papers using hyperref. This script allowed me to add the missing urls and DOI fields to my BibTeX database. As a bonus the script helped me to correct many minor errors that had crept into my BibTeX file over the years (for example, incorrect page numbers and publication years). Now I use the program to automatically update the preprint entries in my database when the papers appear in MathSciNet after they are published.

Options and defaults

Without options, bibupdate uses mrlookup with one AMS lookup at a time, prints compact references to standard output, and writes `updated_file.bib` beside the input `file.bib`. The `--all`, `--arxiv`, `--check`, `--duplicates`, `--keep_fonts` and `--overwrite` switches are off. Field-value wrapping is off (`--wrap 0`).

Updating and checking

`-m, --mrlookup`

Use mrlookup to update bibtex entries (default)

`-M, --mathscinet`

Use mathscinet to update bibtex entries

`-n, --mref`

Use mref to update bibtex entries

These three options select which of the free AMS tools is used to update the BibTeX entries in the database. They are all free services provided by the American Mathematical Society and bibupdate talks to them through the JSON web APIs behind the AMS free tools pages.

By default mrlookup is used. Its advantage is that it allows flexible searching by author, title, year and page numbers, so it can find papers even when they do not yet have an `mrnumber`.

With `-M` the entry is looked up directly by its `mrnumber`. A field containing digits is required: missing, empty or non-numeric MR numbers are skipped, including with `--all` and `--check`.

With `-n` the whole entry is handed to MRef as a free-form reference. Its best matching record is still checked against the local title by bibupdate. Like mrlookup, MRef normally checks only `article`, `book`, `inproceedings` and `incollection` entries without an `mrnumber` field; use `--all` to check all recognised entry types.

`-a, --all`

Update or validate ALL BibTeX entries

By default this switch is off. With mrlookup or MRef, enabling it checks all recognised entry types, including entries with an `mrnumber` field. With `--arxiv`, it also checks entries that already have an `eprint` field. The `-M` tool always requires a usable MR number, even with `--all`.

`-X, --arxiv`

Also add arXiv eprint fields to entries

This switch is off by default. In addition to the usual AMS lookup, query the arXiv for each entry and, when the paper is found there, add `eprint` and `archiveprefix` fields to the BibTeX entry:

```
eprint = {2211.07069},
archiveprefix = {arXiv}
```

bibupdate searches the arXiv using the significant words in the title and only adds these fields when there is a unique paper whose title is a good (fuzzy) match. This option works together with any of the `-m`, `-M` or `-n` lookup tools. The arXiv lookup changes only `eprint` and `archiveprefix`. Entries with an `eprint` field, even an empty one, are skipped unless `-a` or `--check` is used. When such entries are checked, these two fields can be replaced. Entries without a usable title cannot be searched.

bibupdate waits at least 3 seconds after each completed arXiv request before starting the next one. Updating a large database with `-X` can therefore take a while.

`-c, --check, --check1`

Check all entries and report duplicates

This switch is off by default. It prints each entry that would change as a compact reference, using the proposed database fields and retaining any other local fields. The original BibTeX file is not changed and no output bibliography is written. `--check1` is an alias for `--check`. Check mode implies `--all`; it runs arXiv lookups only when `--arxiv` is also supplied. Proposed references remain visible even with `-qq`.

Before network lookups, checks the input file for repeated citation keys and papers listed under different keys. Shared DOI, MR number or arXiv identifiers identify duplicate papers. Matching normalized titles and authors are reported as possible duplicates; capitalization braces, punctuation, spacing and given names versus initials are normalized. Warnings name the keys and source line numbers for review, and remain visible with `--quiet`. Duplicate warnings appear in purple on terminals; log files and redirected output remain plain text.

`--threads N`

Number of concurrent AMS lookups

Use `--threads 2` to overlap two AMS lookups. The default is one thread; the thread count must be a positive integer. This works with mrlookup, MathSciNet and MRef. Entries and log messages retain their original order.

arXiv lookups remain sequential and keep their delay between requests, regardless of the thread count.

Duplicate listing

`-d, --list-duplicates`

List repeated papers and citation keys without changing the bibliography

No network queries, output bibliography or backups are made, even with `--overwrite` or an output filename. Warnings show keys and source lines in purple on terminals, remain visible with `--quiet`, and are suppressed by `-qq`. If no duplicates are found, nothing is printed.

`--list-papers`

List duplicate papers only, including possible title-and-author matches

`--list-keys`

List repeated citation keys only

The same listing-only behaviour and quiet levels apply to all three listing options. Listing options may be combined to show both kinds of duplicates.

Duplicate cleanup

`--duplicates` **(alias** `--deduplicate`**)**

Remove duplicate papers and rename repeated citation keys

All duplicate modes are off by default. This offline cleanup first removes repeated papers, then renames repeated keys among the remaining entries. It makes no AMS or arXiv queries. The output is `updated_file.bib` beside `file.bib`, or the supplied output filename. Use `--overwrite` to replace the input; existing non-empty output files are backed up to `.bak` as usual.

The retained entries keep their original fields and formatting: ignored-field, font-conversion and wrapping options do not apply to cleanup. Cleanup cannot be combined with listing options or `--check`. Other lookup options are ignored.

Changed meaning: `--duplicates` now writes a cleaned bibliography. Use `-d` or `--list-duplicates` for the previous listing-only behaviour.

`--duplicate-papers` **(alias** `--remove-duplicate-papers`**)**

Remove repeated papers, keeping the first occurrence in file order

Papers match through a shared DOI, MR number or arXiv identifier, or through matching normalized titles and authors, as described under `--check`. Connected matches form one group, even when linked through different identifiers. Papers with the same citation key are also compared. Title-and-author matching is heuristic; inspect `--list-papers` before removing entries.

Only the first entry's fields are kept; fields from later entries are not merged. Each removal reports the removed key and the retained key, with source lines. Literal `crossref`, `xref`, `related`, `entryset` and `xdata` references inside the bibliography are updated where the removed key has no surviving occurrence. String macros are preserved without expansion. Citations in external LaTeX files are not changed; use the reported mappings to update citations to removed keys.

`--duplicate-keys` (**alias** `--rename-duplicate-keys`)

Rename repeated keys without deleting papers

Keep the first occurrence of each key. Later occurrences receive a key formed from all authors' surnames and the year, for example `AcharRiche2019`. Editors are used if authors are missing, and `Anonymous` if neither is available. When the key is already used, add `a`, `b`, ..., `z`, `aa`, etc., reserving all original keys and newly generated keys. Each rename is reported with its source line; use `--log` to save the mappings. References to the original repeated key continue to refer to its first surviving occurrence; ambiguous citations need manual review.

Using `--duplicate-papers` together with `--duplicate-keys` is equivalent to `--duplicates`.

Output and formatting

`-o, --overwrite`

Overwrite the existing bibtex file with the updated version

This switch is off by default. Without it, the output is `updated_file.bib` beside the input `file.bib`, or the output filename supplied as the last argument. With `-o`, the input file is replaced and any output filename is ignored. An existing, non-empty output file is copied to its `.bak` file before writing, including when `-o` replaces the input. An existing `.bak` is replaced.

`-i FIELDS, --ignored-fields=FIELDS`

A string of BibTeX fields to ignore when writing the updated file

By default bibupdate removes the following fields from each BibTeX entry:

- `coden`
- `mrreviewer`

The `-i` option replaces this default list. Repeated `-i` options are combined; field names are case-insensitive and separated by spaces. An empty string keeps every field:

```
bibupdate -i "coden fjournal" file.bib   # ignore coden and fjournal
bibupdate -i coden -i fjournal file.bib  # ignore coden and fjournal
bibupdate -i "" file.bib                 # do not ignore any fields
```

`-k, --keep_fonts`

do NOT replace fonts `\Bbb`, `\germ` and `\scr` in titles

This switch is off by default, so supported font commands in titles are replaced. The BibTeX entries generated by mrlookup use `\Bbb`, `\germ` and `\scr` for the `\mathbb`, `\mathfrak` and `\mathscr` fonts. By default, in the *title* fields, these fonts specifications are automatically changed to the following more LaTeX friendly fonts:

- `\Bbb X` --> `\mathbb{X}`
- `\scr X` --> `\mathcal{X}`
- `\germ X` --> `\mathfrak{X}`

Use `-k` to retain the original font commands. This applies both to input titles and to titles retrieved from AMS.

`-w LEN, --wrap LEN`

Wrap bibtex fields to specified width

The default is `0`, which leaves field values unwrapped. A positive `LEN` wraps each field value to that width, with a tab before continuation lines. Field names, braces and indentation add to the total line length, so `LEN` is not a strict limit on complete output lines. Negative values are rejected.

Reports and logging

`-q, --quiet, --quieter`

Print fewer messages

By default, bibupdate prints a wrapped reference for each updated entry and a notice for each attempted AMS lookup that has no unique matching paper, including preprints. With `-q` or `--quiet` it prints brief update notices, missed lookups and network warnings, without separator bars or field-by-field changes. Missed lookups appear in red when the output is a terminal; redirected output and log files contain plain text. Reports wrap at 78 columns. `--quieter` is retained as an alias. Repeat the option (`-qq`) to suppress update notices and lookup warnings. Fatal errors are still reported.

The `--check` option prints proposed references in the compact layout, including when combined with `-q` or `-qq`. Duplicate warnings follow the same warning levels as missed lookups. Use `-l` to send messages to a log file.

`-l LOG, --log LOG`

Log output to file (defaults to stdout)

By default reports and warnings go to standard output. Use `-l` to write them to a file instead; an existing log file is replaced. Fatal errors are also printed to standard error.

Help and version

`-h, --help`

Show a short list of options, grouped by task, and exit.

`-H, --Help`

Print this full description, including grouped options, defaults and known issues, and exit.

`-v, --version`

Print the program version and licence, and exit.

Known issues

- **BibTeX syntax:** the parser uses regular expressions rather than a full BibTeX grammar. Use brace-delimited entries with one field per line and braced values or simple bare words. Multiple fields on one line, quoted values, `#` concatenation, string-macro expansion, and fields containing `@` or `={...}` can be misread or omitted. Inspect the output carefully when using these forms. Duplicate listing and cleanup use a source scanner that preserves quoted and nested values, comments and macros; macros are not expanded for duplicate matching.
- **Title matching:** capitalization braces and apostrophes are removed before searching and comparison. Dollar-delimited mathematics is also removed, which can make distinct titles look alike. Other TeX commands and accents are not fully decoded, so equivalent spellings can still fail to match. A unique fuzzy title match is a heuristic, not a guarantee that the paper is correct.
- **Author names:** mrlookup uses only the first author's surname. A compound surname is preserved in `surname, given names` form; without a comma, only the last word is used. Accents, name particles and alternative spellings can therefore affect searches.
- **Page numbers:** numeric ranges and single numeric pages are supported. Article identifiers and prefixed page numbers are not handled reliably. If no numeric first page is extracted, the title is used instead; for non-book entries the year is then omitted from the query.
- **Ambiguous or missing results:** an AMS entry is updated only when one candidate survives title matching. No match or multiple matches leave it unchanged and produce a missed-lookup notice. Entries without a usable title have less protection against an incorrect match. Skipped entries, such as those with an existing `mrnumber` under the default settings, are not failed lookups and produce no missed-lookup notice.
- **arXiv searches:** searches use significant title words and examine at most ten results. Titles dominated by mathematics or short words can be missed. No unique title match leaves the arXiv fields unchanged; this is normally silent. With `--all` or `--check`, an existing identifier can be replaced by the identifier of the matching title, so review such changes.

- **Network services:** connections can fail, time out or return unexpected data. Requests use a 30-second timeout. Connection errors and timeouts produce warnings, and processing continues. A missed lookup does not prove that the paper is absent from the database.

Rebuilding the documentation

The built-in help in `bibupdate.py` is the source for `README.md` and the marked, generated regions of this manual. The title page, styling and text outside those regions can be edited directly.

Run `python3 setup.py readme` in the source directory to refresh both sources and compile `bibupdate.pdf`. This requires Python, Pandoc, and `latexmk` from a TeX distribution. Use `-no-pdf` to refresh only the sources; `docs` is an alias for `readme`. BibUpdate itself remains a self-contained Python script with no non-standard dependencies.

Feature requests and bug reports

Please make any feature requests and bug reports on the BibUpdate github page:

https://github.com/AndrewMathas/BibUpdate/issues.

Please provide as much detail as possible when submitting requests.

Bug reports must be accompanied by a *minimal working example*, which is a minimal BibTeX entry that demonstrates the problem.

Licence

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